

Measurements and modeling of the phase equilibria for the n-hexane + ethanol + dibutyl ether ternary mixture and its dibutyl ether sub-binary mixtures at 94 kPa

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ABSTRACT

Experimental determinations and theoretical calculations of the phase equilibria for the n-hexane + ethanol + dibutyl ether ternary mixture and the n-hexane + dibutyl ether and ethanol + dibutyl ether binary mixtures are reported in this work at 94 kPa and over the isothermal range 332.3 K to 401.8 K. At the explored conditions, the phase equilibria are characterized by vapor liquid equilibria (VLE) which are measured in an all-glass dynamic VLE cell, whereas the theoretical modeling of the experimental VLE data are carried out by applying classical activity coefficients models (Wilson, and NRTL models), UNIQUAC method, and the molecular based statistical associating fluid theory with variable range using a Mie potential (SAFT-VR Mie) equation of state (EoS).

Considering the experimental determinations, the VLE exhibits positive deviation from the Raoult's law and a zeotropic behavior under the isobaric condition and the isothermal range. The thermodynamical trustworthiness of the measured VLE data is verified by standard consistency test for both binary and ternary mixtures. Based on the statistical evaluation of the dew and bubble points, the Wilson model displays the lower deviations for correlating the VLE of binary mixtures and also to predict the ternary mixture, but the SAFT-VR Mie EoS approach demonstrates a superior capability to model the phase equilibrium of the binary mixtures and to predict the ternary mixture. Finally, it is possible to state that the polarization contribution is not necessary for the mixtures explored in this work.

1. Introduction

Alcohols and ethers have been traditionally used as oxygenate additives in the formulation of gasoline blending. Alcohols (e.g., ethanol and n-butanol) are mixed with fossil fuels to increase their octane rating and also to decrease the emissions of carbon monoxide (CO) in spark-ignition engines, whereas ethers are used as cosolvent for enhancing the miscibility of alcohols within fossil fuel in gasoline mixtures. The MTBE (methyl 1,1-dimethylethyl ether) was initially the selected option because its affinity to fossil fuel and alcohol. However, MTBE has been shown a harmful impact on humans and the environment. In order to reduce the negative effects, several ethers have been tested to replace the MTBE as oxygenate additive. Some of the most explored ethers are the ETBE (ethyl 1,1-dimethylethyl ether), DPE (dipropyl ether), TAME (tert-amyl methyl ether), DIPE (diisopropyl ether), and recently CPME (cyclopentyl methyl ether) has been included as a potential gasoline additive [1]. Excellent reviewers relate to use of oxygenate additives for

fossil fuels are reported by Marsh et al. [2], Halim and Ahsraf-Ali [3], Weber de Menezes et al. [4]. For other works related to the oxygenated gasoline and their thermophysical properties, it is advised some of our previous works (see Refs. [5–9] and references therein). In complement to the studied ethers, dibutyl ether (DBE) has received less attention [10–13] even when its thermophysical properties make it an excellent candidate not only to be used as gasoline additives but also as entrainer to the dehydration of alcohols by heteroazeotropic distillation [14] which together to extractive distillation [15,16] and liquid - liquid extraction with ionic liquids [17] are the most adequate unit operations to produce dry bioalcohols from renewable sources such as waste from vegetables, wood, and forest, as well as food crops, algae, fruits, and cellulose, etc. [18].

Considering these potentiality uses of DBE, the focus of this work is the experimental determination and the theoretical calculations of phase equilibria (vapor-liquid equilibria, VLE) of the ternary mixture conformed by ethanol, n-hexane, and DBE and its DBE binary mixtures.

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Table 1

Technical specification and purity tests for the fluids.

Chemical name	Supplier	Mass fraction purity (as stated by the supplier)	Water Content / ppm ^a	Purification Method	Mass fraction purity (as detected by GC)
Dibutyl ether (DBE)	Aldrich	0.993	102.4	None	0.999
Ethanol	Merck	0.990	100	None	0.999
n-Hexane	Merck	0.990	36.6	None	0.996

^a Uncertainty u is 0.30 wt%.**Table 2**Experimental determination and reference data for the refractive indexes at Na D line (n_D), the mass liquid densities (ρ) of pure fluids evaluated at 298.15 K and 101.3 kPa, and the normal boiling points (T_b) at 101.3 kPa.^{*,**}

Fluid	n_D		$\rho/\text{g cm}^{-3}$		T_b/K	
	Exp.	Lit. ^b	Exp.	Lit. ^b	Exp.	Lit. ^b
DBE	1.39808	1.39662	0.76418	0.76407	413.29	414.86
Ethanol	1.36068	1.35946	0.78506	0.78526	351.44	351.43
n-Hexane	1.37374	1.37308	0.65519	0.65512	341.94	341.82

^{*} Standard uncertainty u are $u(n_D) = 4 \times 10^{-4}$, $u(\rho) = 5 \times 10^{-6} \text{ g cm}^{-3}$, and $u(T) = 0.01 \text{ K}$. ^{**} Lit: NIST-TDE database [19].

Experimental measurements are conducted at an isobaric condition of 94 kPa by using a dynamic Guillestie's VLE device, whereas the theoretical approach is carried out by applying classical activity coefficient models and a molecular based equation of state (EoS). Specifically, three activity coefficient models, namely the NRTL and the Wilson models, as well as the UNIQUAC contribution group have been applied. These models provide a route to correlate the experimental data and then use the results to calculate or estimated other conditions of the phase equilibrium by using the γ - ϕ approach. On the other hand, this work applies the molecular based statistical associating fluid theory using a variable range with the Mie potential (SAFT VR Mie) EoS. This EoS describes the pure fluids and fluid mixtures in terms of the different contributions, such as monomer, formation of chains or rings, the intermolecular association and polar contributions. The main advantage of this latter approach is that the pure fluids and fluid mixtures are described on the terms of their chemical structure, and physical interactions, rather than only energy interactions as activity coefficients or global parameters such as traditional cubic EoS. Additionally, the SAFT approach provides a route to evaluate not only the domain contributions (chain formation, association and polarization) on the phase equilibria, but also a safer extrapolation of the phase equilibria conditions.

2. Experimental section

2.1. Chemical reagents

Anhydrous dibutyl ether was purchased from Aldrich. n-hexane and dried ethanol were acquired from Merck. Technical specification provided by the manufacturers and purity tests (gas chromatography, (GC) and Karl-Fischer titration was used to measure the purity and the water content of the fluids, respectively) are reported in Table 1. Complementarily, the refractive indexes evaluated at Na D line (n_D), the mass liquid densities (ρ), and the normal boiling points (T_b) of the used chemical have been measured to validate the purity of pure fluids. Table 2 collects this information together with the reference experimental values reported on the NIST-TDE database [19].

Table 3

Gas chromatography (GC) column and temperatures of column, detector and injector.

GC column details	Column/ K	Detector/ K	Injector/ K
0.3 cm in diameter and 2 m long, packed with Porapak Q 80/100	523.15	493.15	513.15

Table 4

Polynomial coefficients for gas chromatography (GC) calibration in binary systems and total deviation*.

Binary System	gc ₀	gc ₁	gc ₂	gc ₃	% dev*
Hexane (1) + DBE (3)	1.32053	2.1745	-1.966×10^{-2}	7.965×10^{-5}	0.604
Ethanol (2) + DBE (3)	-0.11007	1.29883	-3.311×10^{-3}	2.986×10^{-6}	0.075

* % dev. = $100 \times \sum (z_i^{\text{cal}} - z_i^{\text{exp}})^2$.

2.2. Gas chromatography, water contents, refractive indexes at Na D line, mass liquid densities, and normal boiling points measurements for pure fluids

Gas chromatography (GC) was used to check the purity of pure fluids and to measure the equilibrium concentration as it will be described later. In this work, a PerkinElmer Clarus 580 gas chromatograph (USA) was used for GC determinations using a thermal conductivity detector (TCD). Table 3 summarizes the column details and the temperature conditions, which provide a good separation.

A Metrohm Karl Fischer titrator 831 (Switzerland) was used to measure the water traces on the pure fluids. A Bellingham + Stanley multiscale refractometer RFM 81 (England) was used to measure the refractive indexes at Na D line of the pure fluids, whereas the mass liquid densities of pure fluids were measured using an Anton Paar DMA 5000 M densimeter (Austria). In the refractometer and densimeter, the temperature was kept constant to within $\pm 0.01 \text{ K}$. Finally, normal boiling points were measured on the same the phase equilibrium cell used for mixture determinations, as it is described below.

2.3. Phase equilibria measurements

Vapor – liquid equilibria (VLE) for the pure fluids and the fluid binary and ternary mixtures explored in this work have been measured in a commercial VLE cell (602 LABODEST cell, Fischer Labor and Verfahrenstechnik, Germany). This cell uses the dynamic experimental technique where both phases (liquid and vapor) are in intense phase contact in a Cottrell pump and then the phases are separated, and their concentration is measured by GC [20]. Specific technical details and experimental procedure have been broadly described in our previous works [5–9].

Briefly, the experimental procedure consists to introduce a mixture into the VLE chamber, and then the pressure is fixed and maintained at the experimental isobaric condition (94 kPa) by using a vacuum pump and nitrogen, and then the mixture in the chamber is heated to its boiling condition. Once the thermal equilibrium condition is reached, 1.0 μL of the condensed vapor and liquid are taken, and their concentration is measured by GC. The GC peak area ratio is converted to mole fraction by using a calibration curve. This curve was built by using samples of known masses of DBE, n-hexane, and ethanol and correlated them to the peak area ratio using a polynomial function.

Specifically, the polynomial function used for binary systems is given by:

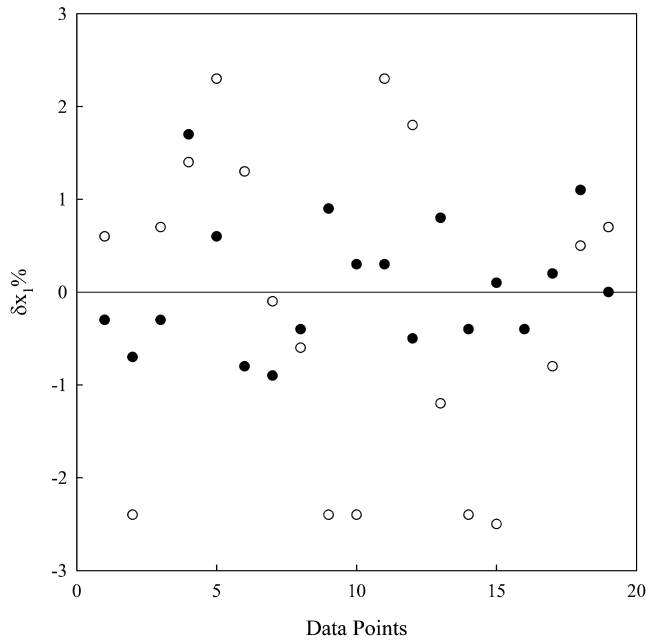


Fig. 1. Residuals of calculated and experimental mole fraction ($\delta x_1\%$) from gas chromatography calibration in binary mixtures. (○) hexane (1) + DBE (3); (●) ethanol (2) + DBE (3).

Table 5

Polynomial coefficients for gas chromatography (GC) calibration in ternary system and total deviation*.

Coefficients	Mole fraction hexane (x1)	Mole fraction ethanol (x2)
gc_{00}	0.0566	0.3071
gc_{01}	1.3213	0.0174
gc_{02}	-0.0039	-0.0004
gc_{03}	0.1061	2.156×10^{-6}
gc_{10}	0.0012	2.2549
gc_{11}	-0.0151	-0.0061
gc_{12}	5.083×10^{-5}	6.943×10^{-6}
gc_{13}	0.0000	0.0000
gc_{20}	0.0000	-0.0216
gc_{21}	8.132×10^{-5}	6.319×10^{-5}
gc_{22}	0.0000	0.0000
gc_{23}	0.0000	0.0000
gc_{30}	-0.1061	8.9893×10^{-5}
gc_{31}	0.0000	0.0000
gc_{32}	0.0000	0.0000
gc_{33}	0.0000	0.0000

$$z_i^{cal} = \sum_{i=0}^3 gc_i An_i^i \quad (1)$$

where z_i^{cal} is the calculated mole fraction for component i , gc_i are the polynomial coefficients, whose numerical values are summarized in Table 4, and An_i is the normalized GC area for component i . For binary systems, $An_i = A_i/(A_i + A_j)$. Eq. (1) covers the mole fraction ranges: $x_1 = (0.051; 0.941)$ for hexane + DBE and $x_1 = (0.052; 0.955)$ for ethanol + DBE. Fig. 1 displays the total deviation of calculated and experimental mole fraction obtained for the two binaries.

For the case of ternary mixture, the polynomial function considers two normalized areas (An_i and An_j): which are related to the CG areas by: $An_i = A_i/(A_i + A_j + A_k)$.

$$z_i^{cal} = \sum_{i,j=0}^3 gc_{ij} An_i^i An_j^j \quad (2)$$

where gc_{ij} are the polynomial coefficients are summarized in Table 5. Eq. (2) reports a total deviation of 1.61%. Eq. (2) covers the mole

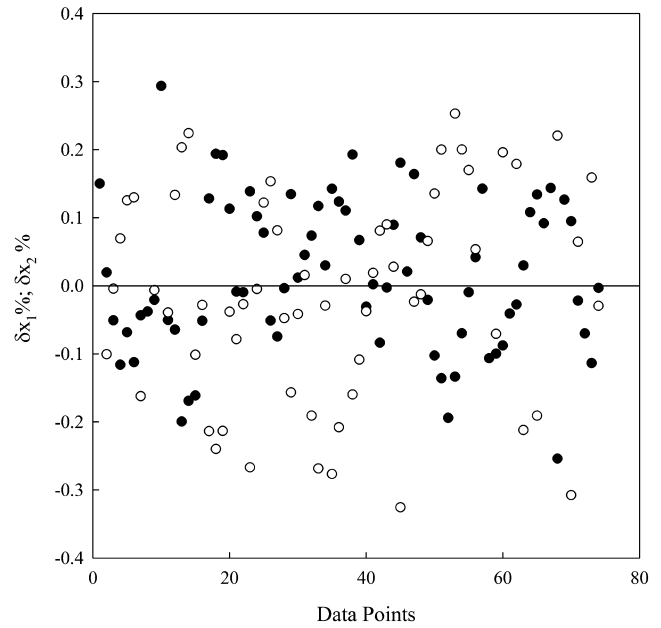


Fig. 2. Residuals of calculated and experimental mole fraction ($\delta x_1\%$, $\delta x_2\%$) from gas chromatography calibration in the ternary mixture hexane (1) + ethanol (2) + DBE (3): (●) $\delta x_1\%$; (○) $\delta x_2\%$

fraction ranges: $x_1 = (0.02; 0.910)$ and $x_2 = (0.034; 0.920)$. Fig. 2 illustrate the corresponding punctual deviations.

Finally, the corresponding measurements uncertainties of the VLE are reported in the footnote of experimental tables. Additionally, Raal and Ramjugernath [21] provide a complete description of this experimental technique.

3. Theoretical section

3.1. Vapor-liquid data treatment and their thermodynamical consistency

Activity coefficients (γ_i) are the fundamental importance to describe the mixture phase behavior, its deviations from the Raoult's law and to evaluate their thermodynamical consistency. In this work, γ_i are calculated from the VLE measurements and the modified Raoult's equation: [22–24].

$$\gamma_i = \frac{y_i P}{x_i P_i^0} \quad (3)$$

In this equation, P represents the absolute pressure, here 94 kPa, P_i^0 is the saturation or vapor pressure of species i , whereas y_i and x_i are the vapor phase and the liquid mole fraction of species i , respectively.

Equation (3) assumes that the vapor phase is described as a perfect or ideal gas, and the liquid phase fugacity is not a function of pressure. This simplification is reasonable here due to the low pressure of the measurement of VLE. Additionally, no further correction of the vapor phase is advised due to the lack of experimental information of the second virial coefficients of DBE and its mixtures, as it was discussed by one of us for the case of ethers. [25]

In this work, the pure fluids saturation pressures (P_i^0) are determinate as a function of temperature (T) employing the same device uses for VLE determinations of mixtures. The experimental $P_i^0(T)$ data are correlated using three parameters (A_i , B_i , C_i) Antoine equation:

$$\log(P_i^0/kPa) = A_i - \frac{B_i}{(T/K) + C_i} \quad (4)$$

The veracity of the measure vapor-liquid equilibria is verified by using traditional thermodynamic consistency tests. [26] Specifically, the thermodynamic consistency of the vapor-liquid equilibria data for the

binary mixtures is evaluated by using Fredenslund et al. [27] approach. In this approach, orthogonal Legendre polynomials are used to describe the Gibbs Excess energy (G^E) of the mixture. This latter function is replaced into the Gibbs-Duhem equation and using least squares regression technique the Legendre parameters are found. According to Fredenslund et al. approach, a VLE datum is considered consistent when the vapor mole fraction difference, $|y_i^{cal} - y_i^{exp}| = \Delta y$, is lower than 0.01. For the case of the ternary mixtures, the thermodynamic consistency is evaluated by applying the Wisniak and Tamir method [28]. In this latter approach, the consistency is evaluated by comparing the local deviation (D) to the maximum deviation (D_{max}) between two experimental points, namely a and b . In this method, the a , b pair of experimental data considered thermodynamically reliable when $D < D_{max}$. D and D_{max} are calculated by the following expressions:

$$D = \sum_{i=1}^3 (x_{ia} - x_{ib})(\ln \gamma_{ia} - \ln \gamma_{ib}) \quad (5.a)$$

$$D_{max} = \sum_{i=1}^3 (x_{ia} + x_{ib}) \left(\frac{1}{x_{ia}} + \frac{1}{y_{ia}} + \frac{1}{x_{ib}} + \frac{1}{y_{ib}} \right) \Delta x \\ + \sum_{i=1}^3 (x_{ia} + x_{ib}) \frac{\Delta P}{P} + 2 \sum_{i=1}^3 |\ln \gamma_{ib} - \ln \gamma_{ia}| \Delta x \\ + \sum_{i=1}^3 (x_{ia} + x_{ib}) B_j \left\{ (T_a + C_j)^{-2} + (T_b + C_j)^{-2} \right\} \Delta T \quad (5.b)$$

where the experimental uncertainties in x_i , T , and P are Δx , ΔT , and ΔP , respectively. In this work, $\Delta x = \pm 0.001$, $\Delta T = \pm 0.1$ K, and $\Delta P = \pm 0.1$ kPa. B_j and C_j are given by Eq. (4).

3.2. Thermodynamical modeling of vapor – Liquid equilibria

The isobaric VLE determinations are modeled in this work by using two approaches: γ - ϕ and ϕ - ϕ . In the γ - ϕ method, the liquid phase is described by using an activity coefficient model (γ), whereas the vapor phase is described as a perfect gas ($\phi = 1$). In the ϕ - ϕ method, both the liquid and vapor phases are modeled by the same equation of state (EoS). These two approaches are discussed in the following sections.

3.2.1. The γ - ϕ approach for vapor – Liquid phase equilibrium

In this method, the phase equilibrium is calculated from Eq. (3), where the vapor phase is modelled as perfect gas, whereas the liquid phase is described by an excess Gibbs energy (G^E) model where the corresponding activity coefficient, γ_b are calculated from its molar partial expression: [22–24]

$$\ln \gamma_i = \frac{1}{RT} \left(\frac{\partial n_T G^E}{\partial n_i} \right)_{T,P,n_j} \quad (6)$$

where R denotes the universal gas constant, T is the equilibrium temperature, n_T represent the total mole in mixture, whereas n_i the mole of species i . P is the equilibrium pressure. In this work, G^E is modelled by using the following three models:

a. The Non-Random-Two-Liquid (NRTL) activity coefficient model: [22–24,29]

$$\frac{G^E}{RT} = \sum_{i=1}^{n_c} x_i \frac{\sum_{j=1}^{n_c} \tau_{ji} G_{ji} x_j}{\sum_{l=1}^{n_c} G_{li} x_l} \quad (7)$$

where τ_{ji} and G_{ji} are given by:

$$\tau_{ji} = A_{ji}/RT \quad ; \quad G_{ji} = \exp(-\tau_{ji} \alpha_{ji}) \quad (8)$$

In Eq. (8), A_{ji} are the energetic parameters of the j - i interaction, and α_{ij} is a symmetric parameter ($\alpha_{ij} = \alpha_{ji}$) that represent the nonrandomness in the mixture. In this work, A_{ij} , A_{ji} , and α_{ij} are found from the phase equilibria experimental data of i - j binary systems.

b. The Wilson activity coefficient model: [22–24,30]

$$\frac{G^E}{RT} = - \sum_{i=1}^{n_c} x_i \ln \left(\sum_{j=1}^{n_c} x_j \Lambda_{ij} \right) \quad (9)$$

where Λ_{ij} term is given by the following expression:

$$\Lambda_{ij} = \frac{V_i^L}{V_j^L} \exp \left(- \frac{A_{ij}}{RT} \right) \quad (10)$$

In Eq. (10), V_i^L and V_j^L are the molar liquid volume of the pure components i and j , respectively. A_{ij} represents the i - j interaction energy. In this work, V_i^L and V_j^L are predicted from the Rackett expression [31], and A_{ij} are found from the VLE data of binary systems.

c. The UNIVERSAL QUAsiChemical (UNIQUAC) activity coefficient model [22–24,32]:

$$\frac{G^E}{RT} = \sum_{i=1}^{n_c} x_i \ln \left(\frac{\Phi_i}{x_i} \right) + \sum_{i=1}^{n_c} q_i x_i \left[5 \ln \left(\frac{\theta_i}{\Phi_i} \right) - \sum_{j=1}^{n_c} \theta_j \tau_{ji} \right] \quad (11)$$

where Φ_i represents the segment fraction of fluid i , θ_i is the area fraction of fluid i , and τ_{ji} are adjustable parameters relate to the characteristic energy. Φ_i , θ_i and τ_{ji} are given by the expression:

$$\Phi_i = \frac{r_i x_i}{\sum_{j=1}^{n_c} r_j x_j} \quad ; \quad \theta_i = \frac{q_i x_i}{\sum_{j=1}^{n_c} q_j x_j} \quad ; \quad \tau_{ij} = \exp \left(- \frac{A_{ij}}{T} \right) \quad (12)$$

where x_i is the mole fraction of the liquid phase for fluid i , whereas the parameter r_i are related to molecular size of fluid i and q_i are related to the external surface areas of the pure fluid i . A_{ji} is the characteristic interaction energy. The numerical values of r_i and q_i are reported in DECHEMA database [33], whereas A_{ji} are found from the phase equilibria experimental data of binary mixtures.

The binary adjustable parameters involve in the G^E models are calculated by minimizing the objective function (OF): [23]

$$OF = (1/N) \sum_{i=1}^N \left(|P_i^{exp} - P_i^{cal}| / P_i^{exp} + |y_i^{exp} - y_i^{cal}| \right)^2 \quad (13)$$

where N is the number of data points, cal stands for calculated quantity whereas exp represents experimental data. In this work, these models are used to model binary mixtures as well as to evaluate the capability for predicting the reported phase equilibria for the ternary mixture only using data from its binary subsystems.

3.2.2. The ϕ - ϕ approach for vapor – Liquid equilibrium

The VLE is calculated from the expressions: [22–24]

$$x_i \hat{\phi}_i^L = y_i \hat{\phi}_i^V \quad ; \quad i = 1, \dots, n_c \quad (14)$$

where x_i and y_i stands for mole fractions of fluid i at the liquid and the vapor phases, respectively. n_c is the number of components. ϕ_i represents the effective fugacity coefficient of species i in the mixture. In this work, ϕ_i is calculated from the expression: [22,24]

$$\ln \hat{\phi}_i = a^R + \left(\frac{\partial a^R}{\partial x_i} \right)_{T, \rho, x_j \neq i} - \sum_{j=1}^{n_c} \left[x_j \left(\frac{\partial a^R}{\partial x_j} \right)_{T, \rho, x_j \neq i} \right] + \left(\rho \frac{\partial a^R}{\partial \rho} - 1 \right) - \ln \left(\rho \frac{\partial a^R}{\partial \rho} \right) \quad (15)$$

where a^R represents the residual Helmholtz energy, ρ represents the molar density of the mixture. In this work, a^R is modeled from the SAFT VR Mie EoS: [34–38]

$$a^R = (a^{MONO} + a^{CHARI} + a^{ASSOC} + a^{POL}) \rho \frac{N_{av}}{\beta} \quad (16)$$

In Eq. (16) $a = A/(N k_B T)$, begin A the Helmholtz energy, N denotes the total number of molecules, k_B is the Boltzmann's constant, and T is the absolute temperature. In Eq. (16), a^{mono} describes the monomer (unbounded) contribution, a^{chari} describes for the formation of chain and/or ring molecules, while a^{assoc} accounts for the intermolecular association contribution, and a^{pol} represents the polar contribution. The corresponding expressions for each contribution are: (see Refs. [34–38] for specific details):

$$a^{MONO} = \left(\sum_{i=1}^{n_c} x_i m_{si} \right) (a^{hs} + \beta a_1 + \beta^2 a_2 + \beta^3 a_3) \quad (17.a)$$

$$a^{CHARI} = - \sum_{i=1}^{n_c} x_i (m_{si} - 1 + \chi_i \eta_i) \ln [g_{ii}^{Mie}(\sigma)] \quad (17.b)$$

$$a^{ASSOC} = \sum_{i=1}^{n_c} x_i \sum_{Ski} \left[\ln X^{Ski} - \frac{X^{Ski}}{2} \right] + \frac{M_{ki}}{2} \quad (17.c)$$

$$a^{POL} = \frac{A_2 / N k_B T}{1 - A_3 / A_2} \quad (17.d)$$

In Eqs. (17), m_{si} symbolizes the spherical segments number that form the chains or rings of fluid i . a^{hs} represents the hard-sphere Helmholtz energy; a_1 to a_3 denote the perturbation terms of hard-sphere; $g_{ii}^{Mie}(\sigma)$ represents the radial distribution function of the Mie potential. In the SAFT-VR-Mie structure monomers interact following the Mie Potential, ϕ^{Mie} : [39]

$$\phi^{Mie}(r) = \epsilon \left[\frac{\lambda_r}{\lambda_r - \lambda_a} \left(\frac{\lambda_r}{\lambda_a} \right)^{\frac{\lambda_a}{\lambda_r - \lambda_a}} \right] \left[\left(\frac{\sigma}{r} \right)^{\lambda_r} - \left(\frac{\sigma}{r} \right)^{\lambda_a} \right] \quad (18)$$

where λ_a and λ_r represent the attraction and repulsion parameters of the intermolecular potential, respectively. ϵ denotes the potential well depth (energy scale), σ represents the effective segment diameter (length scale), r denotes the center to center distance of the interacting segments.

In Eq. (17.c), the a^{assoc} contribution which is described by the number of association sites of species i (M_{ki}), and the non-bonded site fraction of sites Ski , X^{Ski} , which is defined as:

$$X^{Ski} = \left[1 + N_{av} \sum_{j=1}^{n_c} \sum_{Bwj} \rho_j X^{Bwj} \Delta^{SkiBwj} \right]^{-1} \quad (19)$$

where N_{av} is Avogadro's constant, i , and j are the species in the mixture, k_i are the association types for species i , and w_j are the association types for species j , which includes the species i . Δ^{SkiBwj} is the association force between Ski and Bwj association types. Δ^{SkiBwj} is described as a function of the site-site energy of interaction (ϵ^{AB}); the association range (r_c^{AB}), and the distance between the associating site and its corresponding segment center (r_d^{AB}). In this work, the association scheme is represented by three parameters, namely the number of bipolar (B), positive (P), and negative (N) sites. This formulation allows us

to work with cross-associating fluid mixtures where the site balance is solved numerically. [40–42]

Finally, a^{POL} (Eq. (17.d)) is a Padé approximation for the Helmholtz energy for polarization contribution. In this expression, the terms A_2 and A_3 represent the second- and third-order of the Padé approximation to describe the polarization of species. A_2 and A_3 are a function of the potential energy parameter, ϵ , the size parameter, σ the dipolar moment of the fluid, μ , and the number of polar sites, n_p .

From the previous expressions, pure fluids are characterized by up to eleven parameters: m_s , λ_r , λ_a , ϵ , σ , χ , ϵ^{AB} , r_c^{AB} , r_d^{AB} , μ and n_p . Fortunately, some of these parameters are known from the molecule structure and type. For example, χ , ϵ^{AB} , r_c^{AB} , r_d^{AB} , μ and n_p are zero for pure non-polar, non-associating chain fluids; μ is known for polar fluids; r_d^{AB} is usually fixed at 0.4σ . Additionally, the Mie potential is conformal, meaning that infinite λ_r - λ_a pairs can describe the same macroscopic fluid properties, proving that the selected λ_r - λ_a pair gives the same van der Waals constant value. In this scenario, it is common to fix one of the exponents and leaving the other as an adjustable parameter. In this work, $\lambda_a = 6$. This selected value is based on the London dispersion theory. Considering the above observations one is left with seven adjustable parameters: m_s , λ_r , ϵ , σ , ϵ^{AB} , r_c^{AB} , and n_p to fit the most general case. The numerical values of the molecular parameters used in this work have been taken from Cripwell et al. [38] for ethanol. For n-hexane and dibutyl ether, their numerical values are obtained from the minimization of the objective function (OF):

$$OF = \frac{1}{N} \sum_{i=1}^N \left(\left| \frac{P_i^{0,cal} - P_i^{0,exp}}{P_i^{0,exp}} \right| + \left| \frac{\rho_i^{L,cal} - \rho_i^{L,exp}}{\rho_i^{L,exp}} \right| \right) \quad (20)$$

Table 6

Experimental determination of the saturation pressures (P^0) as a function of temperature (T) for DBE*.

T/K	P^0 /kPa
375.08	30
379.59	35
383.59	40
387.19	45
390.45	50
393.45	55
396.27	60
398.70	65
401.10	70
403.33	75
405.49	80
407.52	85
409.42	90
411.01	95
413.17	101.32

* Standard uncertainty u are $u(T) = 0.05$ K and $u(P) = 0.03$ kPa.

Table 7

The Antoine constants (A_i , B_i and C_i) for pure fluids*.

Fluid	A_i	B_i	C_i	Temperature range/K
DBE	8.54371	3329.5600	95.97533	375.10 to 413.20
Ethanol ^a	7.16873	1551.6458	-50.90503	323.51 to 351.68
n-Hexane ^b	6.25283	1310.6332	-33.20660	307.67 to 342.08

* The constants values are from: ^a Mejía et al. [44]. ^b Mejía et al. [45].

Table 8

Molecular parameters using in SAFT-VR Mie EoS for pure fluids*.

Fluid	m_s	$\sigma/\text{\AA}$	$\epsilon/k_B/K$	λ_r	$\epsilon^{AB}/k_B/K$	r_c^{AB}/σ	[B, P, N]	μ/D	n_p
DBE ^a	3.288	4.179	340.84	16.871	—	—	[0, 0, 1]	—	—
DBE (Pol) ^{a,b}	3.276	4.183	336.78	16.562	—	—	[0, 0, 1]	1.16	3.995
Ethanol ^c	1.772	3.559	224.50	11.319	3018.05	0.3547	[1, 0, 1]	—	—
Ethanol (Pol) ^c	2.010	3.398	201.59	10.733	2815.52	0.3864	[1, 0, 1]	1.69	0.9106
n-Hexane ^a	1.967	4.547	377.60	18.411	—	—	—	—	—

The values of λ_a and r_d^{AB}/σ are fixed to 6 and 0.4, respectively. ^a. The numerical values are obtained from the experimental data reported here and NIST-TDE database [19]; ^b. the values of μ are fixed and their values are taken from NIST-TDE database [19]; ^c. numerical values taken from Cripwell et al. [38].

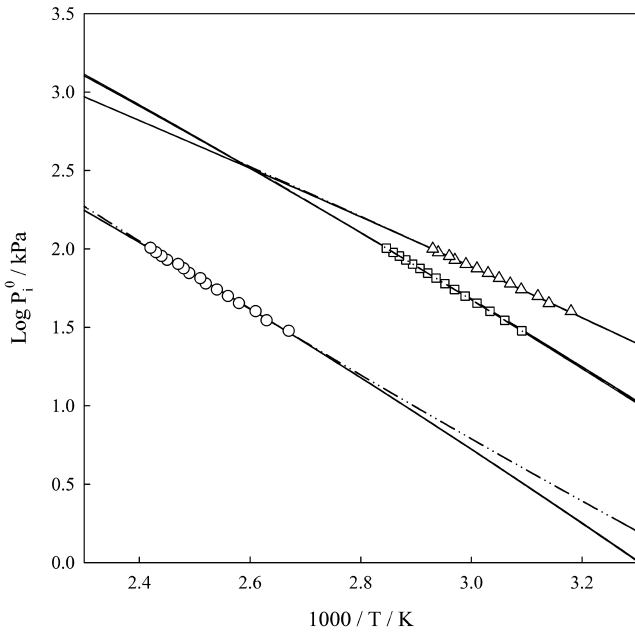


Fig. 3. Vapor pressure for pure fluids in $(P - T)$. (— · —) Antoine equation (see Eq. (4) and Table 7); (—) SAFT-VR Mie EoS (see Table 8); Experimental data: (□) Ethanol (Mejía et al. [44]); (○) DBE (this work); (△) n-hexane (Mejía et al. [45]).

In Eq. (20), P_i^0 and ρ_i^l denote the saturation pressure, and the density at liquid state of species i , respectively. The superscript exp represents experimental value while cal stands for calculated from the EoS. N symbolizes the number of experimental points.

For the case of fluid mixtures, the SAFT-VR Mie EoS used the following cross parameters for size (σ_{ij}), energy (ϵ_{ij}) and Mie exponents ($\lambda_{r,ij}$ and $\lambda_{a,ij}$). These unlike parameters are calculated by the following combining rules:

$$\sigma_{ij} = \frac{\sigma_{ii} + \sigma_{jj}}{2} \quad (21.a)$$

$$\epsilon_{ij} = \sqrt{\epsilon_{ii}\epsilon_{jj}} \frac{\sqrt{\sigma_{ii}^3\sigma_{jj}^3}}{\sigma_{ij}^3} (1 - k_{ij}) \quad (21.b)$$

$$(\lambda_{k,ij} - 3) = \sqrt{(\lambda_{k,ii} - 3)(\lambda_{k,jj} - 3)} \quad ; \quad k = r, a \quad (21.c)$$

In Eq. (21.b), k_{ij} is the interaction binary parameter, which is found by fitting the EoS prediction to experimental phase equilibria data of binary mixtures.

Finally, it is important to state that n-hexane and DBE do not self-associate ($\epsilon^{AB} = 0$) but they associate with ethanol. A simple approach is to take the unlike association energy as the half of the self-associating molecule and fit the association geometry to experimental data. As r_d^{AB} is usually fixed to 0.4σ , this implies that only r_c^{ABij} is optimized. Specify details concerned with the SAFT VR Mie calculations (e.g., pure fluids saturation pressure, optimization of the involved parameters, and calculation of phase equilibria in mixtures) have been recently described by some of us. [43]

4. Results and discussions

4.1. Pure fluids

The saturation or vapor pressures (P_i^0) were measured as a function of temperature (T) using the same device as that for experimental determination of the phase equilibria data in mixtures. For DBE, the saturation pressure data summarized in Table 6.

The values reported in Table 6 display a mean absolute percentage deviation (MAD) of 2% to previous vapor pressure data reported on NIST – TDR [19] and DECHEMA [33] databases. The Antoine equation (See Eq. (4)) accurately correlates the $P^0(T)$ with an MAD of 0.22% using the numerical values reported in Table 7. For ethanol and n-hexane, $P^0(T)$ have been measured in our previous works [44,45] and their values have been verified in this work. The corresponding Antoine constants for ethanol and n-hexane are included in Table 7.

The vapor pressure data reported here together with the corresponding values reported in the NIST-TDE database [19] for both saturation vapor pressure and liquid mass density have been used in Eq. (20) to obtain the EoS parameters for DBE and n-hexane pure fluids. For ethanol, the corresponding parameters are directly taken from Cripwell et al. [38]. Table 8 summarizes the pure fluid numerical parameters, where two models (i.e., with and without polarization (pol) contribution) are included for ethanol and DBE.

Fig. 3 displays the saturation pressure as a function of temperature for the pure fluids used here. This figure includes experimental data, Antoine correlation, and SAFT VR Mie results.

From Fig. 3, it is possible to conclude that both Antoine correlation and SAFT-VR Mie results are able to describe the vapor pressure of the pure fluids in the explored temperature range. It is also noted that the SAFT VR Mie results are similar with or without polarization contribution. Table 9 collects the MAD obtained from the Antoine correlation

Table 9

Summary of the absolute average deviation (AAD) for vapor pressure*.

Fluid	Antoine	SAFT-VR Mie without polarization	SAFT-VR Mie with polarization
DBE	0.22%	0.44%	0.44%
Ethanol	0.02%	0.45%	0.35%
n-Hexane	0.05%	0.35%	—

* % AAD $P = (100/N) \sum_{i=1}^N |P_i^{\text{exp}} - P_i^{\text{cal}}| / P_i^{\text{exp}}$. N number of data points; exp represents experimental data while cal stands for calculated quantity. The experimental data are taken from: DBE (this work), ethanol (Mejía et al. [44]), and n-hexane (Mejía et al. [45]).

Table 10Experimental determinations of VLE for hexane (1) + DBE (3) mixture at $P = 94.0$ kPa ^{a,b}.

T/K	x_1	y_1	γ_1	γ_3
410.76	0.000	0.000		1.000
396.97	0.074	0.397	1.128	0.994
392.25	0.106	0.500	1.102	0.993
388.84	0.139	0.567	1.038	0.997
384.80	0.182	0.642	0.990	0.991
383.93	0.190	0.660	0.996	0.977
381.60	0.205	0.686	1.015	0.996
381.40	0.208	0.699	1.024	0.964
374.60	0.286	0.780	0.989	0.984
366.19	0.393	0.865	0.999	0.956
362.15	0.445	0.898	1.023	0.914
359.78	0.486	0.912	1.016	0.934
356.93	0.533	0.931	1.025	0.890
352.73	0.615	0.953	1.029	0.871
349.83	0.674	0.965	1.037	0.845
348.20	0.725	0.969	1.017	0.942
344.71	0.793	0.980	1.046	0.931
344.11	0.831	0.981	1.019	1.097
339.83	0.947	0.994	1.038	1.232
339.45	1.000	1.000	1.000	

^a T is temperature; x_1 is the liquid phase mole fraction of hexane (1); y_1 is the vapor phase mole fraction of hexane (1); γ_1 and γ_3 are the activity coefficients of hexane (1) and DBE (3), respectively. ^b Standard uncertainties u are $u(T) = 0.01$ K, $u(x_1) = u(y_1) = 0.001$, and $u(P) = 0.03$ kPa.

and the SAFT-VR Mie EoS with and without polarization for DBE and ethanol.

4.2. Binary mixtures

The measurements of the vapor – liquid equilibrium (VLE) for n-

Table 11Experimental determinations of VLE for ethanol (2) + DBE (3) mixture at $P = 94.0$ kPa ^{a,b}.

T/K	x_2	y_2	γ_2	γ_3
410.76	0.000	0.000		1.000
377.36	0.088	0.689	2.841	0.991
372.37	0.114	0.751	2.822	0.972
367.61	0.148	0.795	2.722	0.984
362.44	0.213	0.837	2.396	1.019
360.41	0.256	0.851	2.188	1.061
357.63	0.336	0.871	1.890	1.145
356.64	0.379	0.880	1.761	1.181
355.27	0.447	0.890	1.592	1.274
354.45	0.501	0.895	1.473	1.393
353.68	0.561	0.904	1.369	1.495
353.09	0.615	0.910	1.286	1.636
352.58	0.668	0.915	1.216	1.820
351.88	0.723	0.925	1.167	1.977
351.54	0.771	0.930	1.115	2.268
349.55	1.000	1.000	1.000	

^a T is temperature; x_2 is the liquid phase mole fraction of ethanol (2); y_2 is the vapor phase mole fraction of ethanol (2); γ_2 and γ_3 are the activity coefficients of ethanol (2) and DBE (3), respectively. ^b Standard uncertainties u are $u(T) = 0.01$ K, $u(x_1) = u(y_1) = 0.001$, and $u(P) = 0.03$ kPa.

Table 12Numerical values of the Legendre parameters (L_1 , L_2 , and L_3) and statistics used for consistency of n-hexane (1) + DBE (3) mixture and ethanol (2) + DBE (3) mixture at 94.0 kPa.

Binary	L_1	L_2	L_3	$\delta\gamma^a$	$\delta P^a/\text{kPa}$
(1) + (3)	0.10356	0.02863	0.17577	0.008	0.88
(2) + (3)	1.40153	0.18919	-0.01268	0.004	0.91

^a Average of absolute deviation: $\delta\theta = (1/N) \sum_{i=1}^N |\theta_i^{\text{exp}} - \theta_i^{\text{cal}}|$, where $\theta = y$ (vapor mole fraction) or P (pressure), N represent the experimental data number.

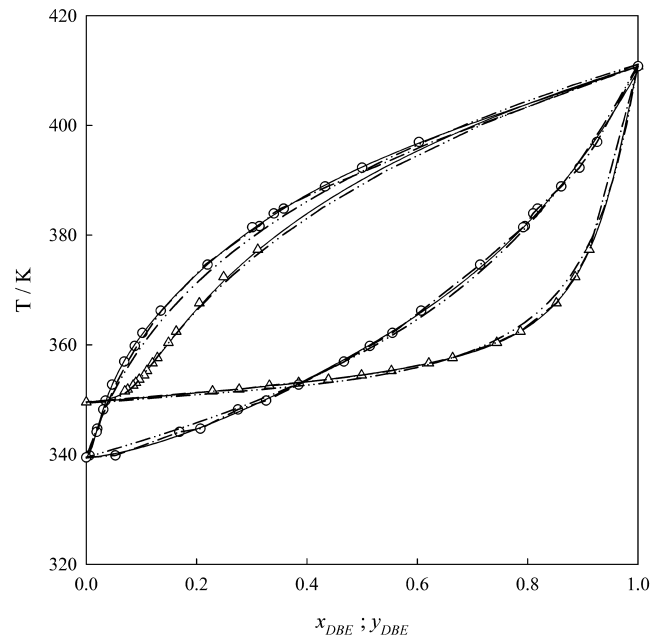


Fig. 4. Phase Equilibrium diagram (T , x_{DBE} , y_{DBE}) for n-hexane (1) + DBE (3) mixture and ethanol (2) + DBE (3) mixture at 94.00 kPa. Experimental data: $\circ$, n-hexane (1) + DBE (3); $\triangle$: ethanol (2) + DBE (3). —, three-degree Legendre polynomial prediction used in thermodynamical consistency test (c.f., Table 12); - - -, Wilson model (c.f., Table 13); - · -, SAFT VR Mie EoS (c.f., Table 14).

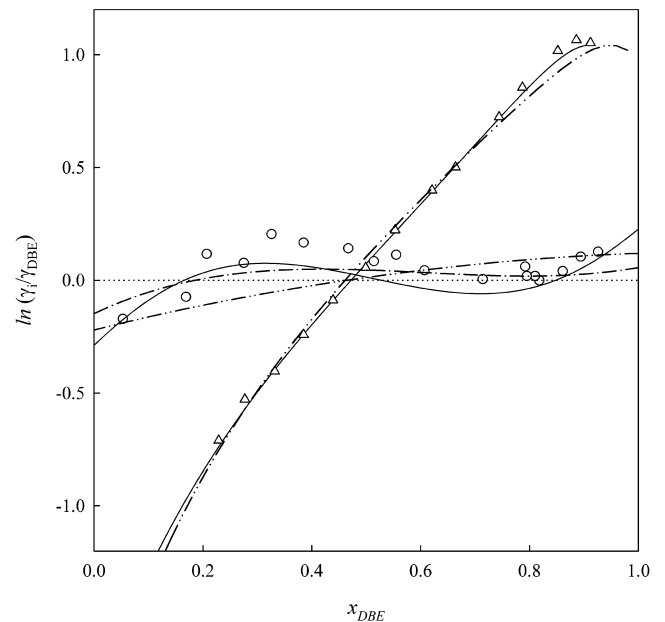


Fig. 5. $\ln(\gamma_i/\gamma_{\text{DBE}})$ vs. x_{DBE} for the n-hexane (1) + DBE (3) mixture and ethanol (2) + DBE (3) mixture at 94.0 kPa. Experimental data: $\circ$, n-hexane (1) + DBE (3); $\triangle$: ethanol (2) + DBE (3). —, three-degree Legendre polynomial prediction used in thermodynamical consistency test (c.f., Table 12). - - -, Wilson model (c.f., Table 13); - · -, SAFT VR Mie EoS (c.f., Table 14).

hexane (1) + DBE (3) and ethanol (2) + DBE (3) at $P = 94$ kPa are collected in Tables 10 and 11, respectively. These tables include the equilibrium temperature T , the liquid x_i and vapor y_i phase mole fractions of component i , and the calculated activity coefficients, γ_i , from Eq. (3).

From the measure VLE data reported in Tables 10 and 11, it is possible to conclude that both binary systems exhibit zeotropic

Table 13

Gibbs energy Excess (G^E) parameters and the corresponding statistics deviation for the n-hexane (1) + ethanol (2), n-hexane (1) + DBE (3) and ethanol (2) + DBE (3) binary systems at 94.0 kPa.^{a,b}

Model/ system	$A_{ij}/J\ mol^{-1}$	$A_{ji}/J\ mol^{-1}$	α_{ij}	Bubble-point pressures		Dew-point pressures	
				ΔP (%) ^f	$100 \times \Delta y_i^g$	ΔP (%) ^f	$100 \times \Delta x_i^g$
NRTL							
(1) + (2)	5574.670	3815.070	0.45	0.56	0.70	1.22	3.50
(1) + (3)	−1502.022	2045.034	0.40	2.04	0.62	2.62	1.54
(2) + (3)	3557.283	1705.301	0.40	0.79	0.33	1.20	0.96
Wilson							
(1) + (2)	1296.13	8867.43		0.57	0.50	1.80	1.90
(1) + (3)	−1652.027	3472.108		1.70	0.57	2.27	1.08
(2) + (3)	5084.445	598.798		0.76	0.32	1.40	0.86
UNIQUAC ^c							
(1) + (2)	3517.770	−616.070		2.80	2.2	1.32	4.70
(1) + (3)	−1594.175	1931.857		1.96	0.53	2.19	1.27
(2) + (3)	−680.058	3295.812		0.86	0.36	1.29	1.47

^a The numerical values of G^E parameters were obtained from Eq. (13) and the experimental information reported here (Tables 10 and 11), except to n-hexane (1) + ethanol (2) mixture where the parameters are reported by Mejía et al. [6]. ^c the numerical values of the volume (r) and area (q) UNIQUAC molecular parameters are taken from DECHEMA [33]: $r_1 = 4.4998$, $r_2 = 2.1055$, $r_3 = 6.0925$, $q_1 = 3.8560$, $q_2 = 1.972$, $q_3 = 5.176$. ^f % $\Delta P = (100/N) \sum_i^N |P_i^{exp} - P_i^{cal}| / P_i^{exp}$. ^g % $\Delta \theta = (1/N) \sum_i^N |\theta_i^{exp} - \theta_i^{cal}|$ with θ_i : x_i or y_i .

Table 14

SAFT VR Mie EoS interaction parameters (k_{ij}) and their prediction statistics for the binary mixtures of n-hexane (1) + ethanol (2), n-hexane (1) + DBE (3) and ethanol (2) + DBE (3) at 94.0 kPa.^a

System	k_{ij}	$r_c^{ABij}/\text{\AA}$	% ΔP^b	$100 \times \Delta y^b$	% ΔP^b	$100 \times \Delta x^b$
(1) + (2)	0.0109	–	0.99	0.82	0.62	1.19
(1) + (2 – pol)	-0.0104	–	1.04	0.87	0.52	1.50
(1) + (3)	0.0066	–	2.09	1.18	4.36	2.22
(1) + (3 – pol)	0.0047	–	2.10	1.19	4.38	2.24
(2) + (3)	0.0205	2.305	2.65	1.09	1.15	2.05
(2) + (3 – pol)	0.0191	2.302	2.67	1.09	1.17	1.99
(2 – pol) + (3 – pol)	0.0086	2.364	2.63	1.06	1.23	1.78

^a The numerical values of the interaction parameters (k_{ij}) were obtained by minimization using the experimental information reported here (Tables 10 and 11). For the case of (1) + (2) binary system, the VLE data are reported by Sinor and Weber [46]. ^b see footnote of Table 13.

behavior. The n-hexane (1) + DBE (3) binary mixture behaves almost ideally ($\gamma_i \approx 1$). These results are in agreement to the isothermal data reported at 308.15 K and over a pressure range from 3.66 kPa to 28.09 kPa by Marsh et al. [10], and at 298.15 K and over 1.74 kPa to 19.21 kPa by Lepori et al. [11].

For the case of ethanol (2) + DBE (3) binary mixture, this mixture displays positive deviation from Raoult's law ($\gamma_i > 1$) Comparing the measured results to previous experimental data determinations by Han et al. [12] at 333.15 K and Gmehling [13] at 323.15 K, 338.15 K and 352.15 K and over a pressure range from 3.38 kPa to 104.4 kPa, it is possible to conclude that the isobaric measurements reported here are in agreement with the VLE measured at isothermal conditions, that is the mixture is zeotropic with a positive deviation from Raoult's law.

As we described in the Section 3.1, any report of new experimental VLE data need to include the corresponding thermodynamic consistency test. In this work, the experimental measurements of the phase equilibrium reported in Tables 10 and 11 were correlated by applying the Legendre polynomials, where three coefficients were needed to describe the experimental VLE data [26] and the fulfil the consistency criteria.

Table 15

Statistics deviation from PR + MHV(Wilson) model for the prediction of VLE of binary and ternary mixtures at 94.0 kPa.

Mixture	Bubble-point pressures		Dew-point pressures	
	% ΔP^a	$100 \Delta y_i^b$	% ΔP^a	$100 \Delta x_i^b$
Hexane (1) + DBE (3)	1.79	1.55	3.22	2.00
Ethanol (2) + DBE (3)	2.22	1.22	1.78	1.18

^a % $\Delta P = (100/N) \sum_i^N |P_i^{exp} - P_i^{cal}| / P_i^{exp}$. ^b % $\Delta \theta = (1/N) \sum_i^N |\theta_i^{exp} - \theta_i^{cal}|$ with θ_i : x_i or y_i .

Table 16

Experimental determinations of the VLE for n-hexane (1) + ethanol (2) + DBE (3) mixture at $P = 94.0$ kPa.^{a,b}

T/K	x_1	x_2	y_1	y_2	γ_1	γ_2	γ_3
332.24	0.609	0.286	0.635	0.359	1.323	2.617	0.932
332.32	0.714	0.177	0.662	0.333	1.171	3.903	0.851
332.54	0.500	0.400	0.617	0.376	1.550	1.940	1.025
333.33	0.388	0.511	0.590	0.402	1.858	1.563	1.226
333.44	0.808	0.081	0.707	0.287	1.065	6.978	0.820
334.87	0.279	0.613	0.551	0.439	2.296	1.328	1.355
335.26	0.599	0.188	0.622	0.365	1.189	3.539	0.854
335.48	0.495	0.290	0.584	0.401	1.343	2.499	0.923
336.60	0.382	0.399	0.539	0.443	1.547	1.910	1.088
336.89	0.698	0.081	0.686	0.300	1.068	6.248	0.857
337.14	0.182	0.706	0.481	0.504	2.848	1.199	1.695
337.85	0.281	0.509	0.501	0.480	1.877	1.536	1.157
339.21	0.490	0.166	0.574	0.402	1.180	3.728	0.852
339.44	0.388	0.282	0.514	0.460	1.325	2.481	0.957
340.57	0.280	0.396	0.458	0.513	1.577	1.878	1.003
340.68	0.177	0.602	0.420	0.554	2.280	1.326	1.350
340.96	0.588	0.079	0.663	0.311	1.073	5.638	0.886
342.04	0.081	0.796	0.335	0.642	3.809	1.100	1.943
342.87	0.384	0.178	0.508	0.453	1.188	3.358	0.905
343.27	0.177	0.496	0.366	0.596	1.830	1.553	1.204
343.50	0.283	0.280	0.432	0.526	1.342	2.410	0.962
344.52	0.084	0.695	0.285	0.677	2.887	1.196	1.673
345.51	0.178	0.386	0.333	0.616	1.545	1.881	1.091
345.93	0.091	0.597	0.260	0.692	2.317	1.341	1.428
346.14	0.476	0.074	0.628	0.326	1.070	5.049	0.918
347.37	0.285	0.164	0.437	0.506	1.196	3.380	0.904
347.52	0.093	0.498	0.227	0.713	1.898	1.552	1.279
347.93	0.180	0.274	0.322	0.613	1.372	2.384	1.006
349.39	0.093	0.396	0.217	0.711	1.705	1.807	1.137
350.56	0.371	0.080	0.561	0.372	1.071	4.466	0.942
350.97	0.181	0.181	0.326	0.592	1.258	3.093	0.972
351.54	0.094	0.287	0.201	0.712	1.473	2.292	1.041
354.51	0.096	0.179	0.200	0.690	1.315	3.170	1.008
357.24	0.270	0.074	0.497	0.403	1.071	4.026	0.916
357.95	0.181	0.100	0.360	0.524	1.132	3.795	0.936
363.80	0.094	0.091	0.235	0.599	1.204	3.829	0.956

^a T denotes temperature; x_1 and x_2 are the liquid phase mole fraction for hexane (1) and ethanol (2), respectively. y_1 and y_2 are the vapor phase mole fraction for hexane (1) and ethanol (2), respectively. γ_1 and γ_2 are the activity coefficients of hexane (1) and ethanol (2), respectively. ^b Standard uncertainties u are $u(T) = 0.01$ K, $u(x_1) = u(y_1) = 0.001$, and $u(P) = 0.03$ kPa.

Table 12 summarizes the Legendre coefficients and the statistics to evaluate the consistency.

Figs. 4 and 5 show the experimental measurements of the VLE for hexane (1) + DBE (3) mixture and ethanol (2) + DBE (3) mixture and their activity coefficients at 94 kPa (see Tables 10 and 11), respectively. These figures include the results from Legendre polynomial (c.f. Table 12). From these figures, one can visualize a good agreement between the measurements of the VLE and the calculated activity coefficients with the three terms Legendre polynomial model used in thermodynamical consistency for both binary systems.

The numerical VLE data collected in Tables 10 and 11 are also employed to parametrize the Wilson and the NRTL activity coefficient models as well as the UNIQUAC group contribution model by using Eq. (13), under bubble point pressure as well as dew point pressure

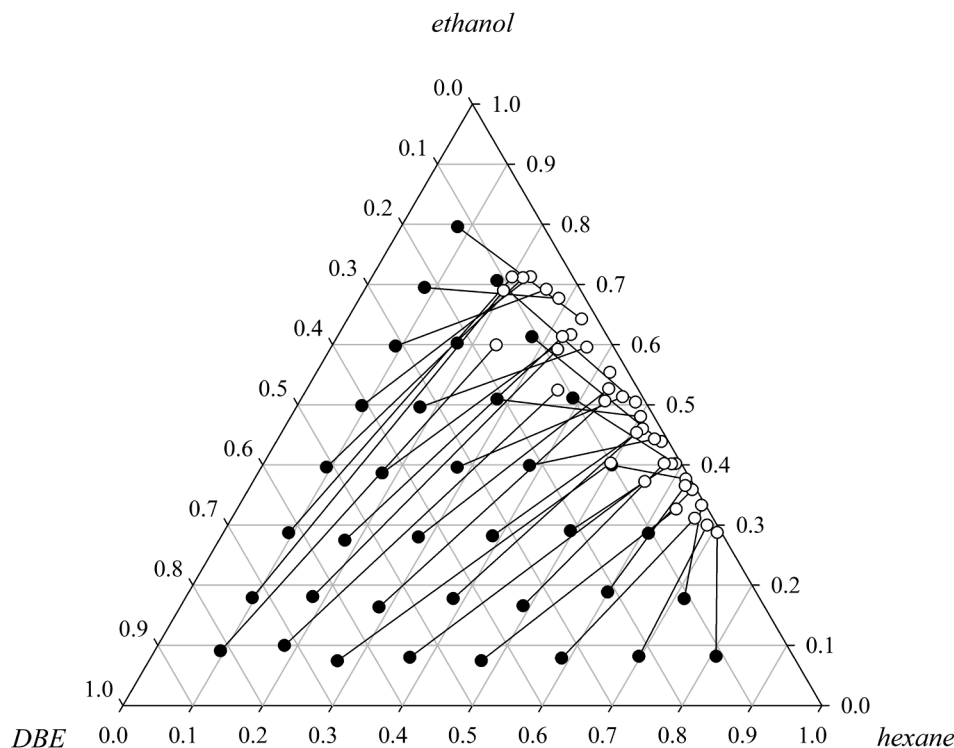


Fig. 6. Experimental vapor – liquid equilibria for the n-hexane (1) + ethanol (2) + DBE (3) ternary mixture at 94.0 kPa. (○) vapor phase; (●) liquid phase; (—) tie lines.

calculations. The numerical values of corresponding parameters for each model and their statistical deviation under bubble point pressure and dew point pressure calculations are collected in Table 13. In the latter table, the corresponding numerical values of the activity coefficient models for the n-hexane (1) + ethanol (2) binary mixture are also included for complementary information and further discussion relates to the capability to predict the ternary (n-hexane (1) + ethanol (2) + DBE (3)) mixture from its binary subsystems.

Based on the low deviations results of both bubble- point pressure and dew point pressure reported in Table 13, one can conclude that VLE data for the binary mixtures can be well correlated using all considered activity coefficient models, but it is observed that Wilson activity coefficient model exhibits the lowest deviation.

In complement to the activity coefficient models, the VLE for the two binaries reported here have been calculated from the molecular based EoS. In this case, the EoS model uses an interaction parameter (k_{ij}) which is fitting from experimental VLE data. For the case of n-hexane + ethanol mixture, k_{ij} is obtained using the experimental data reported by Sinor and Weber [46]. Table 14 collects the values of k_{ij} and the corresponding statistical deviations (bubble and dew point pressures) for each binary mixture. This table also includes the results with and without polarization.

From the results, it is possible to conclude that the polarization term does not significantly impact the results for the binary mixtures explored here. However, it is worthy to observe that SAFT VR Mie EoS describes with a lower deviation the experimental data than traditional activity coefficient models. In order to compare the performance of the SAFT VR Mie EoS to other EoSs, the Peng – Robinson EoS [47] with the Stryjek-Vera's thermal cohesion function [48] and extended to mixtures using the modified Huron – Vidal mixing rule (MHV) [49] have been selected. The latter mixing rule uses an activity coefficient model. In this work we selected the Wilson model, which was the model that represents better the experimental data. (see Table 13). In Supplementary information, the main expressions of the model are summarized. For a complete discussion of the latter approach, the reader is redirected to Ref. [8].

Table 17

Statistics for predicting n-hexane (1) + ethanol (2) + DBE (3) ternary mixture at 94.0 kPa from the binary Gibbs excess (G^E), SAFT VR Mie and PR + MHV (Wilson) models*

Model	Bubble-point pressures			Dew-point pressures		
	ΔP (%) ^c	$100 \times \Delta y_1$ ^d	$100 \times \Delta y_2$ ^d	ΔP (%) ^c	$100 \times \Delta x_1$ ^d	$100 \times \Delta x_2$ ^d
NRTL ^a	3.02	0.98	1.14	1.60	1.08	3.23
Wilson ^a	3.11	0.95	1.12	1.32	0.72	3.28
UNIQUAC ^a	14.53	4.75	4.03	7.85	5.22	4.45
SAFT VR Mie ^b						
(1) + (2) + (3)	1.81	1.27	1.49	2.96	1.13	3.59
(1) + (2) + (3-Pol)	1.80	1.26	1.49	3.10	0.82	2.60
(1) + (2-Pol) + (3-Pol)	1.89	1.20	1.42	3.35	0.90	2.68
PR + MHV(Wilson)						
(1) + (2) + (3)	2.73	0.88	1.05	1.16	0.96	3.10

* The values are reported in: ^a Table 13; ^b Table 14; ^c See footnote of Table 13.

Table 15 summarizes the statistics deviation from the latter approach.

4.3. Ternary mixture

Table 16 collects the experimental determination of the phase equilibria (VLE) for n-hexane (1) + ethanol (2) + DBE (3) ternary mixture at 94.0 kPa. Table 16 includes the equilibrium temperature T , the liquid x_i and vapor y_i phase mole fractions of component i , and the calculate activity coefficients, γ_i , from Eq. (3).

From these results, one can conclude that this non azeotropic ternary system displays positive deviation from Raoult's law (or ideal behavior) with $G^E > 1$. Fig. 6 displays the equilibria mole fraction together with the tie lines, where it is possible to observe there are no azeotropic behavior.

In order to validate the VLE measurements reported in Table 16, the Wisniak and Tamir consistency test was applied. [26]. According to the

results, the lower D_{max} value was 0.029, which always is bigger the calculate D values. Therefore, the reported data of ternary VLE data are thermodynamic consistent.

Finally, the reliability to predict the experimental VLE ternary data from binary contribution was evaluated by only considering binary contributions on the NRTL, Wilson, UNIQUAC, SAFT VR Mie and PR + MHV(Wilson) models (see Tables 13–15). The corresponding are collected in Table 17.

According to the results collected in Table 17, the Wilson activity coefficients model exhibits the lower deviation for correlating the ternary system at 94 kPa. However, it is possible to observe that the SAFT-VR Mie EoS is superior to the activity coefficients and PR + MHV (Wilson) in its capability to predict the ternary mixture. Additionally, it is possible to conclude that the polarization contribution is not significant for the mixtures explored here.

5. Conclusions

In this work, consistent experimental measurements and theoretical calculations of the phase vapor–liquid equilibria at 94 kPa for the ternary system n-hexane + ethanol + DBE and its DBE binary mixtures are reported. On the base of experimental measurements and theoretical results, the phase equilibrium data of the sub-binary mixtures and the ternary mixture exhibit zeotropic behaviour. The sub-binary mixture of n-hexane + DBE displays an ideal behavior whereas the ethanol + DBE binary mixture and the ternary mixture displays positive deviation from the Raoult's law.

The phase vapor–liquid equilibrium data of the binary mixtures were well-modeled from Wilson, NRTL, and UNIQUAC activity coefficients model, and the ternary mixture was predicted with good agreement only using binary parameters. Considering the statistical deviation results, the Wilson activity coefficient model is the most appropriated model for correlating the phase equilibria for both binaries and ternary mixtures.

The theoretical calculations based on SAFT VR Mie EoS of vapor–liquid phase equilibrium are in very good agreement with the reported measurements of the binary mixtures and it is also able to predict the ternary mixture with lower deviation than the activity coefficients models. Finally, the polarization contribution on the EoS model does not show significant improvements.

CRedit authorship contribution statement

Marcela Cartes: Methodology, Data curation, Investigation, Writing - review & editing. **Gustavo Chaparro:** Methodology, Software, Formal analysis, Writing - review & editing. **Andrés Mejía:** Conceptualization, Methodology, Formal analysis, Data curation, Software, Writing - original draft, Writing - review & editing, Validation, Investigation, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Theoretical modeling of the vapor – liquid phase equilibria from Peng-Robinson equation of state and modified Huron – Vidal mixing

rule. Supplementary data to this article can be found online at <https://doi.org/10.1016/j.fuel.2021.120361>.

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